

Dariush Ghaemi

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Research Interests

Safety in Autonomous Driving, Sensor Fusion, Deep Learning, Machine Learning, Signal Processing, Pattern Recognition

Education

Sharif University of Technology

Sep. 2024 - Aug. 2026

MSc in Electrical and Electronics Engineering (Integrated Circuit Design)

Current GPA: 16.84/20

Sharif University of Technology

Sep. 2019 - Mar. 2024

BSc in Electrical Engineering

Current GPA: 17.76/20

Allameh Helli High School

Sep. 2016 - Jun. 2019

Diploma in Physics and Mathematics

GPA: 19.63/20

Research Projects

Sensor Fusion in Autonomous Driving

Sep. 2024 - Present

MS Project, Advisor: Prof. A. Fotowat-Ahmady, Prof. B. H. Khalaj

- Built Carla from source code
- Designed and implemented custom maps using RoadRunner and Unreal Engine
- Implemented simple Autonomous Driving Algorithms.

Quality Control of a Screw Production Line using Computer Vision

Jul. 2022 - May 2023

BS Project, Advisor: Prof. M. Hashemi

- Designed and implemented a screw detection algorithm (edge detection), then extracted screw features to check them.
- Designed and implemented a template matching algorithm to check the validity of the screw
- Ran the algorithms real-time using a camera

Awards and Honors

Apr. 2026	Ranked 10th in National University Entrance Exam , for PhD Degree in Electronics and Digital Systems
Jul. 2024	Experimental Committe Assistant in the 54th International Physics Olympiad
May 2024	Ranked 17th in National University Entrance Exam , for M.Sc. Degree in Electronics and Digital Systems
Jul. 2023	Ranked among the top 25% of the class of 2023 , B.Sc. in Electrical Engineering program at Sharif University of Technology
Sep. 2018	Golden Medal in the National Physics Olympiad (Exams included classical mechanics, thermodynamics and data analysis.)

Work Experience

Teaching Assistantship

Sep. 2021 - Aug. 2026

Sharif University of Technology

- Engineering Mathematics (Fall 2021)
- Probability and Statistics (Fall 2022)
- Computer Vision Lab (Fall 2023)
- Electronics 2 (Fall 2025)
- Electronics 2 Lab (Fall 2025)

Computer Vision Engineer

Mar. 2022 - May 2023

RetinaTech

- Quality Control using computer vision in industry

High School Teacher and Curriculum Advisor

Jun. 2019 - Sep. 2021

Preparing students for IPHO and National Physics Olympiad

- Allameh Helli High School
- Mandegar Alborz High School

Skills

Python Programming	OpenCV, CVXPY, PyTorch, Keras, TensorFlow
Programming	Python, C, C++, Java/JavaScript, MATLAB, Verilog, VHDL
Software	Carla, Unreal Engine, RoadRunner, L ^A T _E X, Simulink
Languages	Persian (Native), English (C1-Advanced)

Selected Courses

- Machine Learning
- Convex Optimization
- Digital Signal Processing
- Linear Algebra and Applications in Learning and Optimization
- Object-Oriented Programming
- CMOS1 (graduate course)
- Neural Modeling (graduate course)
- Computer Vision (graduate course)
- Advanced Computer Vision Lab
- Deep Learning (graduate course)
- Parallel Programming(graduate course)

References

- Prof. A. Fotowat-Ahmady
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- Prof. B. H. Khalaj
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